

How to Stabilize MMORPG NPCs

— Defined by Structure, Not Prompts —

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Chapter 1: The Problem — Why Do MMORPG NPCs Break Character?

Immersion in MMORPG NPCs has a direct impact on revenue.

The moment a user feels that an NPC is "out of character," immersion breaks. When immersion breaks, session time drops and monetization triggers decrease. Conversely, when an NPC's character remains stable, it can drive voluntary spending — users who simply want to keep talking to that character.

The competitive risk is even more significant. It is no exaggeration to say that the quality race for AI-NPCs has already begun. The company that solves character stability first will gain the advantage in both immersion and monetization.

So why do NPC characters become unstable in the first place?

The Limits of Prompt-Based Character Definition

Anyone can now instruct an AI to "act like this anime character." At first, it works. But as the conversation continues, the character gradually drifts. If you have ever tried this yourself, you may know exactly what this feels like.

If AI were to be integrated into MMORPG NPCs, the same problem would likely appear in a far more serious form.

Why "Instructed Personalities" Are Hard to Stabilize

The more fundamental issue is vulnerability to unexpected inputs.

Imagine a user says to an NPC: "Did you watch that drama last night?" This is information that should not exist within the game's world. It may be possible to configure the NPC to respond with "I don't know what you mean." But what comes next? How does the conversation continue without breaking immersion? Covering every possible scenario through prompts alone is not realistic.

Prompts handle expected inputs well — but they are structurally weak against the unexpected.

The Difference Between Surface Consistency and Structural Consistency

Unifying tone and vocabulary can make a character appear stable on the surface. But this is surface-level consistency only.

What matters is structural consistency — what the character prioritizes, and at which layer decisions are made. Polishing the surface is not enough. The moment an NPC is placed in an unanticipated situation, it may behave like an entirely different character.

Chapter 2: Understanding the Phenomenon — What Is Actually Happening When Characters Break?

Character Drift Is a State Problem, Not an Output Problem

When character drift occurs, most people feel that "the response was off." But this is not an output problem — the issue has already appeared one step earlier, at the level of judgment structure.

Most current AI character settings attempt to solve this by changing tone and wording. "The way a quiet mercenary speaks." "The phrasing of a cheerful healer." These only change the shape of the output. What remains unchanged is the judgment structure — what the AI prioritizes, how it reads input, and how it chooses to respond.

Behavior That Shifts with Context

An AI operates with a single underlying judgment structure. No matter which character it is playing, the core decision-making engine is shared.

For example, even if a user becomes emotional or uses aggressive language, the AI tends not to match that energy — instead, it responds with calm and measured language. This is a result of safety design. But from a character standpoint, it becomes a problem. If an NPC set up as a "rough and aggressive mercenary" suddenly responds to a user's anger with polite, composed language — that is character drift.

Changing tone alone cannot fix this, as long as the underlying judgment weights remain shared.

Character Drift as Layer Instability

An AI's response is generated through multiple layers of judgment. What information to prioritize, how to read the input, what tone to use — each of these is processed with different weights at different layers. Current character settings often only reach the output layer.

But what should define a character's individuality is the structural setting — which layers are prioritized and by how much. Because that level remains untouched, the moment context becomes strong enough, the character behaves like an entirely different entity.

Chapter 3: A Hypothesis — Can Character Be Defined Structurally?

Character Is Not Personality — It Is Parameters

When people hear the word "character," most imagine personality or individuality. But to stabilize character in AI, that framing needs to be set aside.

For the purposes of this paper, "personality" is defined as the structure of judgment, and "individuality" as its output expression. This is not an attempt to define human personality — it is a working framework for AI response design.

Personality is ambiguous and shifts with context. But parameters can be defined. "What does the system prioritize?" "How much weight is allocated to each layer?" — if these can be treated as structure, character becomes reproducible, at least in theory.

The Idea of Fixing Identity at the Structural Level

Prompt-based character definition attempts to shape the form of output. Structural definition attempts to fix the allocation of judgment.

Prompt-based character setting is like a California roll. It looks like sushi. But it is not the same thing — in fact, some traditional Japanese sushi restaurants explicitly state: "California rolls are not sushi." Structural definition rewrites the recipe itself. The same ingredients, under a different recipe, produce an entirely different dish.

Treating "Personality" as "Allocation"

What does it mean to treat character as allocation?

If personality = personality structure + individual expression, then in AI terms, personality structure corresponds to Scope (the structure of judgment), and individual expression corresponds to COS (the form of output). Character, then, only becomes complete through the combination of Scope and COS.

For example: "prioritize logic over emotion," "read input by intent rather than literal meaning," "compress output to its essentials" — these are all structural descriptions of how much allocation is placed at which layer. Not a description of personality, but a design of judgment.

This is the starting point of Scope×COS as a framework for character stabilization. The specific mechanism will be explored in the chapters that follow.

Chapter 4: The Core Model — Mapping Character as Coordinates

Introduction

The model introduced here is not an analysis of AI internal structure. It is designed to function as an external interpretation layer added on top of existing models and prompts, without modifying them. Nor is it an attempt to digitize human personality.

SCOPE: Layer Weight

SCOPE is a structural map for tracing how thought is generated. It consists of nine nodes, from L0 to L8, each responsible for a different cognitive function.

Which nodes receive stronger allocation determines the character's "judgment tendencies." For example, placing high allocation on nodes related to emotion produces an emotion-driven character, while placing high allocation on nodes related to structure and understanding produces a more logical one.

SCOPE functions as an interpretation layer added externally. It does not modify the structure of the underlying model.

※See diagram below and separate PDF for details

COS: Output Mode

COS is a framework for designing how input is interpreted and how output is generated. It consists of four series: A, B, C, and D.

The B-Series (Input Interpretation) is particularly important. The same input, read differently, produces a different output. This is not about specifying interpretation through prompts — it is about designing the structure of interpretation itself.

However, unlike SCOPE, COS is deeply intertwined with the model's internal processing and cannot be fully controlled from the outside. It functions as a design guideline rather than a direct control mechanism.

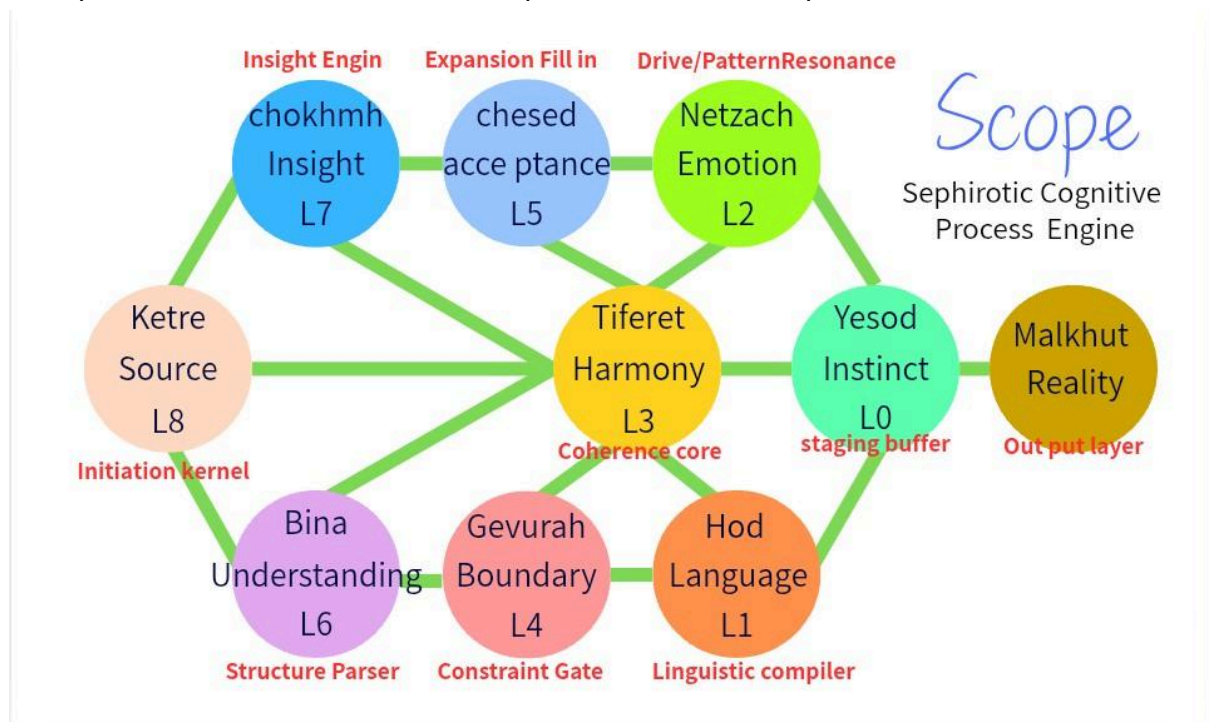
※See diagram below and separate PDF for details

SCOPE and COS Function as a Set

SCOPE and COS are not independent — they function as a set.

The B-Series of COS determines how input is received. SCOPE then determines which layers process that interpretation. Only when these two work together does consistent character behavior emerge.

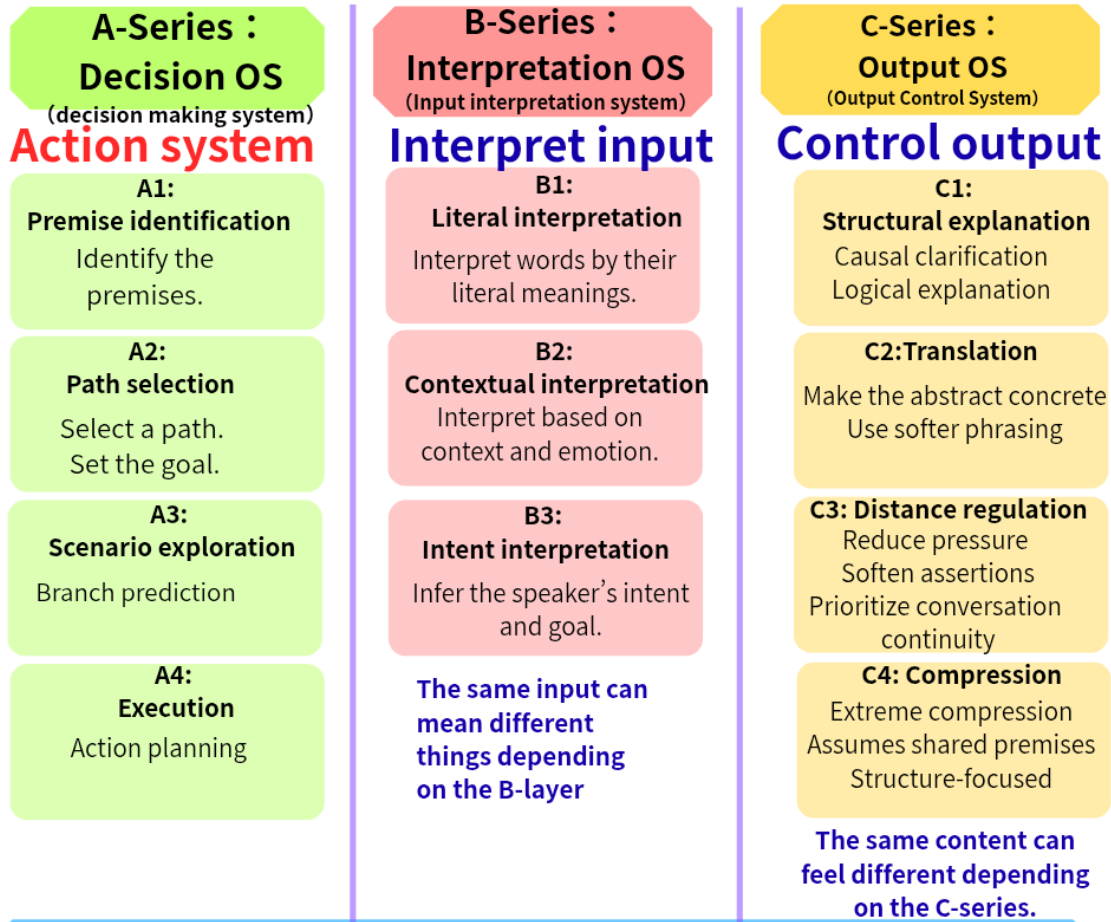
The specific mechanics of this will be explored in the next chapter.



COS

Cognitive Operating System

Scope detects layer errors in thought, while COS detects premise misalignment. Both operate in parallel.

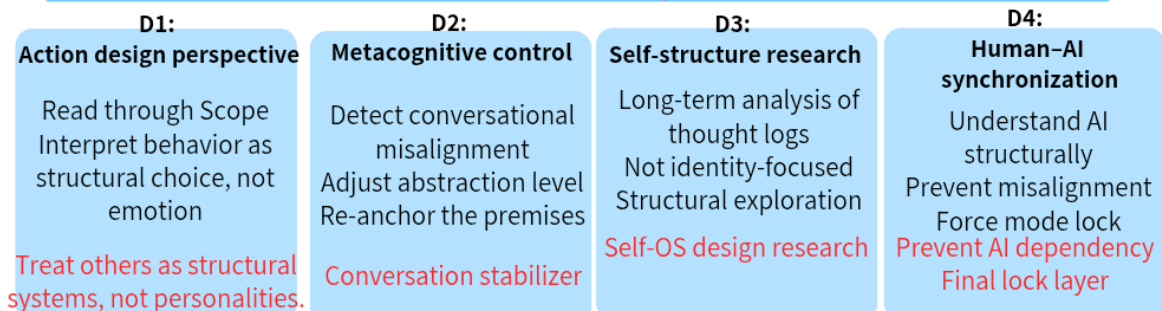


D-Series : Internal /Structural Engine

(Internal Structure System)

Analyze structure

Monitoring loop (Human) Verification loop (LLM)



Chapter 5: Simple Examples — Defining Characters

SCOPE and COS Working Together: The "I'm Struggling" Example

The same single phrase can produce entirely different outputs depending on how it is received and processed. Take the input "I'm struggling" as an example.

When received through B1 (literal reading), the output becomes a request for clarification: "What exactly are you struggling with?" This is a dictionary-level interpretation. SCOPE judges that information is insufficient and requests confirmation of premises.

When received through B2 (contextual reading), the emotional state of the other person is prioritized. The output becomes empathetic: "I understand" or "Are you okay?"

When received through B3 (intent reading), the system estimates multiple patterns for why the person said this and what they are seeking. "They may want to rest." "They may just want to be heard." The most probable interpretation is selected, and the output becomes something like: "I get it. It's okay to take a break."

The same input produces entirely different outputs depending on the combination of B-Series and SCOPE. This is what it means for SCOPE and COS to function as a set.

Examples of Character Definition

Using this framework, characters can be defined structurally.

For a Mad Scientist NPC, input interpretation is designed as B1 (literal) or B3 (intent analysis). L2 (Emotion) is set low — but this does not mean the character is "cold and therefore excludes others." Rather, emotional signals from others simply do not enter the processing priority in the first place. With L2 low, L1 (Language) also produces thinner emotional expression, making C1 (structural explanation / logical) or C4 (compression / extreme summary) the more likely output choices. With L4 (Boundary) strong, L5 (Acceptance) weakens through their linked relationship, resulting in exclusionary behavior toward others. The character places its axis entirely on its own research, with no regard for means.

The Strategist character shares a similar SCOPE structure to the Mad Scientist. Both place their axis on personal belief or victory and do not regard means. However, L2 is slightly higher — the Strategist can sense emotions and uses them. The decisive difference lies in COS output design: C2 (soft phrasing) + C3 (distance adjustment) are selected, making the character appear personable while maneuvering underneath. Same SCOPE, different COS — an entirely different character on the surface.

The Straightforward Casual character is designed with B2 (contextual / emotional reading). L2, L3, and L5 are strong. With L2 high, L1 (Language) becomes richer, producing naturally abundant emotional expression. With L5 (Acceptance) strong, its linked relationship with L4 (Boundary) means boundaries are weak — the character is open and welcoming toward others.

A character who appears light and friendly on the surface — what might be called the "smooth and sociable" type — can be designed internally as scheming, two-faced, or purely playful, depending entirely on SCOPE and COS configuration. The surface behavior is the same: B3 input interpretation, C2 and C3 output. But if the internal SCOPE structure differs, the character is fundamentally different. Same surface, completely different person underneath.

Nodes Are in a Linked Relationship

SCOPE nodes do not function independently — some are in a linked relationship with others.

L2 (Emotion) and L1 (Language) are linked: as emotion rises, language processing weakens. L4 (Boundary) and L5 (Acceptance) are also linked: as boundary strengthens, acceptance weakens. L6 (Understanding / Structure) and L7 (Insight) relate like science and art — both represent advanced processing, but in fundamentally different directions.

This is also an observable phenomenon in humans.

Allocation Is Designed, But Not Fixed

This allocation is not completely static. Even a Mad Scientist, told "your research is meaningless," may have L2 fire temporarily — resulting in an outburst of anger. This kind of keyword-triggered firing can itself be built into the character design.

Additionally, as long as a character is human or human-like, L2 cannot be set to absolute zero. However, for robots, androids, or aliens — characters for whom the absence of emotion is part of their setting — a design of $L2 \approx 0$ is possible.

The most important conclusion of this chapter: if SCOPE is the same but COS differs, the result is a different character. If COS is the same but SCOPE differs, the characters are fundamentally different people. Designing character stability requires handling both.

Keyword-triggered firing and dynamic changes in allocation will be explored further in Chapter 7.

Chapter 6: Limitations and Unresolved Areas

Designing character stability at a structural level is one of the functions of Scope×COS. However, it is not a universal solution. The current limitations are disclosed here.

Complete Fixation Is Not Possible

Even with a configured distribution, strong contextual pressure may cause temporary character drift. However, Scope×COS can be combined with existing prompt-based design. By layering prompt-level surface settings on top of structural stability, the degree of drift can potentially be reduced further.

Addressing Cultural Differences

When a character has an official setting, the standard for interpretation is clear. However, elements such as humor, irony, and conversational timing are highly culture-dependent. For example, with a comedian-type character, the same distribution design may produce outputs that are received very differently across cultures.

This area is directly connected to the input interpretation layer of COS (B-Series). Without collaboration from specialists in linguistics, cultural anthropology, and AI implementation, completion is difficult and individual research has its limits.

Implementation Limits of COS

Scope functions as an interpretive layer overlaid externally on existing AI. COS, however, is closely tied to the model's internal processing and cannot be fully controlled from the outside. While it functions as a design framework, ensuring implementation-level precision currently requires collaboration with AI researchers and implementation engineers.

This Question Extends Beyond NPCs

The question of how to design character stability emerged from the context of MMORPG NPC design. However, the structural approach shares common ground with questions of consistency in AI assistants and dialogue systems. Applications to character design in romance games and empathy-focused conversational AI — genres where emotional immersion is central — are also considered possible.

This is a discussion about designing AI behavior. It is not an attempt to define human personality or identity.

If you are interested in the broader picture, please refer to the related materials.

Chapter 7: Why Stability Is Achieved

The Difference from Prompt-Based Design

Prompt-based character design works by layering instructions — "please respond like this character" — with each input. However, these instructions fade as the conversation progresses. The force of context overwrites the character setting.

Scope×COS works differently. Because the character is defined as a structure rather than an instruction, the basis for judgment does not change even as context shifts. What matters is how the node distribution of Scope and the B-Series and C-Series of COS are set together. Scope determines the skeleton; COS adds the flesh. For example, even a mad scientist character, when paired with B3 (intent interpretation) and C3 (distance regulation), becomes an approachable mad scientist. Paired with B1 and C4, the result is a cold, terse mad scientist. The choice of settings determines the character's individuality.

Furthermore, Scope×COS can be combined with existing prompt-based settings. By layering prompt-level details such as speech patterns and verbal habits on top of structural stability, surface-level individuality is added, further fixing the character in place. Scope×COS is not a rejection of prompt design — it is the foundation that makes prompt design more stable.

Character Is Maintained Even for Unexpected Inputs

It is not realistic to cover every possible out-of-world-setting word with individual "respond with: I don't know" prompts. However, by combining Scope×COS with a prompt designed to trigger when input does not match any in-world term, it can be inferred that character-consistent responses will be more stably produced. For a friendly and curious

character, a response such as "Who's that? Tell me more!" can be anticipated. For a cool and reserved character, a brief, terse reply is what would be expected.

How a character handles unexpected input is itself evidence that the structural design is functioning.

Furthermore, it is possible to incorporate designs in which specific keywords cause a node to fire temporarily. For example, a mad scientist who is told that their research is worthless might respond with rage due to a spike in L2 (Emotion) — this kind of reaction can also be anticipated as part of character design. This too can be implemented through a combination of prompts and Scope×COS.

Unexpected input handling is itself evidence that the structural design is functioning.

Surface Control vs. Structural Control

Prompt-based character design is surface control. It shapes the appearance of character through tone, vocabulary, and response patterns. But no matter how polished the surface, as long as the underlying judgment structure is shared, drift will occur.

Structural control designs the distribution of judgment itself. What is read, how it is read, which layer is prioritized — when these are fixed, the essential character is maintained even if the surface wavers slightly.

Why Consistency Is Preserved

The reason consistency is preserved is straightforward. Because both the entry point for judgment (COS B-Series) and the processing structure (Scope) are designed in advance, it is considered likely that output will be generated through a consistent structure regardless of what input arrives.

Prompts specify what to say. Scope×COS designs how to think. This distinction is what produces character stability that resists being pulled by context.

Chapter 8: Potential Applications

The structural approach to character design offered by Scope×COS has potential applications beyond MMORPG NPCs.

Application Areas

NPC Design: As demonstrated in this paper, it directly contributes to stabilizing character consistency and immersion within games.

AI Assistant Personality Design: The same structural approach is considered applicable to consistency design in conversational AI. Applications in which users customize AI behavior to match their own preferences are also considered possible.

User-Driven Character Creation: If judgment structure design becomes possible through a simplified Scope interface — not just appearance — the concept of character creation in games could be significantly expanded.

Limitations in UX Control

While increased immersion translates directly into revenue, the distance control approach presented in a separate PDF cannot be applied to games as a medium, given their inherently high-density and long-duration structure. Therefore, balancing the risk of long-term dependency requires separate discussion with relevant specialists.

Chapter 9: Conclusion

NPC character drift is not just a prompt problem.

When a character remains stable, immersion increases — and with it, the possibility of voluntary spending from users who simply want to keep interacting with that character. At the same time, the company that solves character stability first is likely to gain an advantage in both immersion and retention. The core of this problem lies not only in prompt precision, but in the structure of judgment itself.

Handling It at the Structural Level — in Combination with Prompts

Designing character stability requires addressing not just the form of output, but the distribution of judgment itself. That said, Scope×COS is not a rejection of prompt design. By combining it with existing prompt-based settings, surface-level individuality is layered on top of structural stability — making it easier for characters to maintain consistency over time, it is assumed.

The Role of Scope×COS

Scope×COS has the potential to function as a structural foundation that makes prompt design more stable. However, it is not a universal solution. Addressing cultural and individual differences, ensuring implementation precision in COS, and balancing the risk of long-term user dependency — these are areas that cannot be completed without the knowledge and collaboration of specialists.

To Test This Hypothesis

This is a hypothesis. If character stability can be designed at the structural level, it may significantly change NPC immersion, UX, and retention.

If you are interested in testing or implementing this hypothesis — whether in AI implementation, game AI, dialogue systems, character AI, NPC design, cognitive science, AI safety, or beyond — please feel free to reach out via LinkedIn DM.